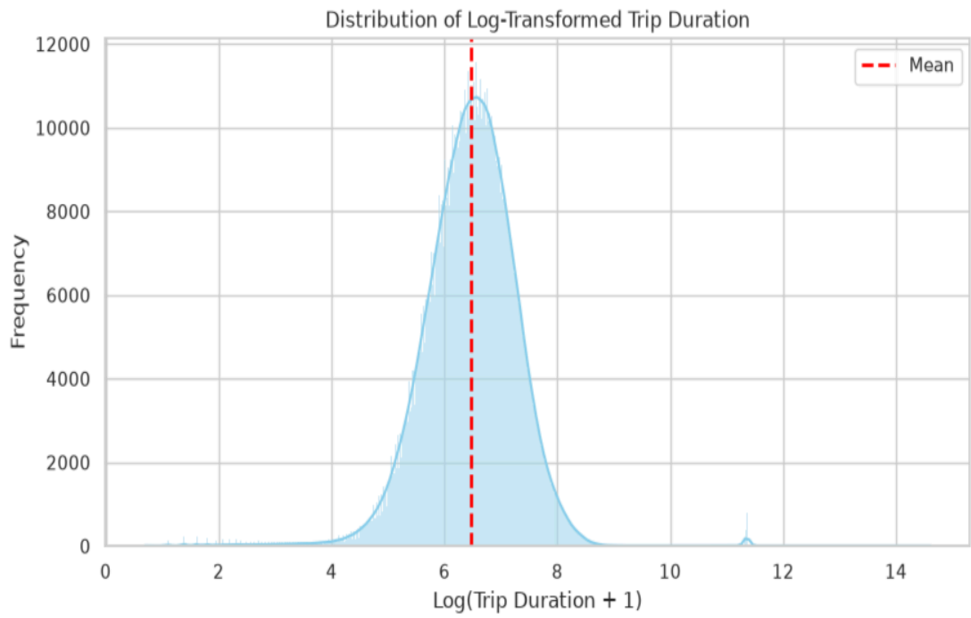
Trip Duration Prediction Project Report

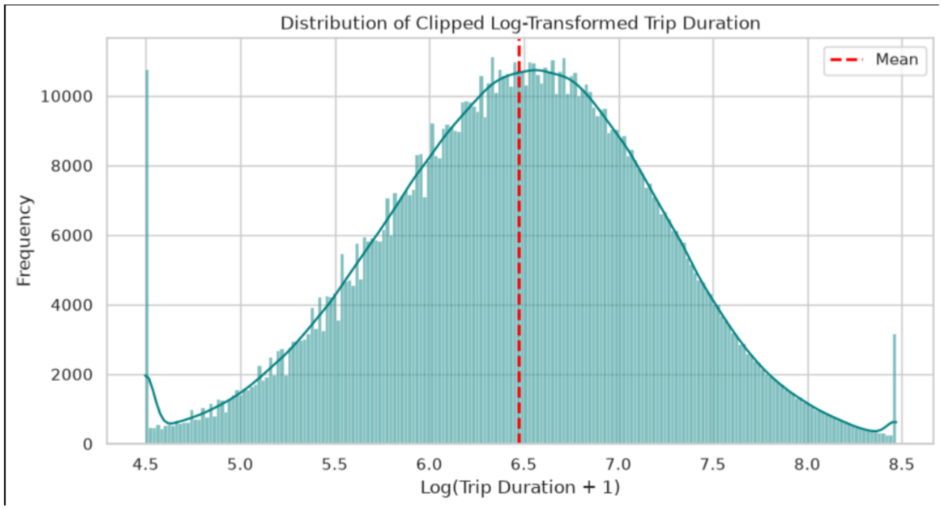
1 Introduction

The NYC taxi trip duration prediction is a competition on Kaggle to build a model to predict the total taxi trip duration in NewYork City. The primary dataset has some information as pickup time , geo information to start and end trips and other information.

2 shape of data

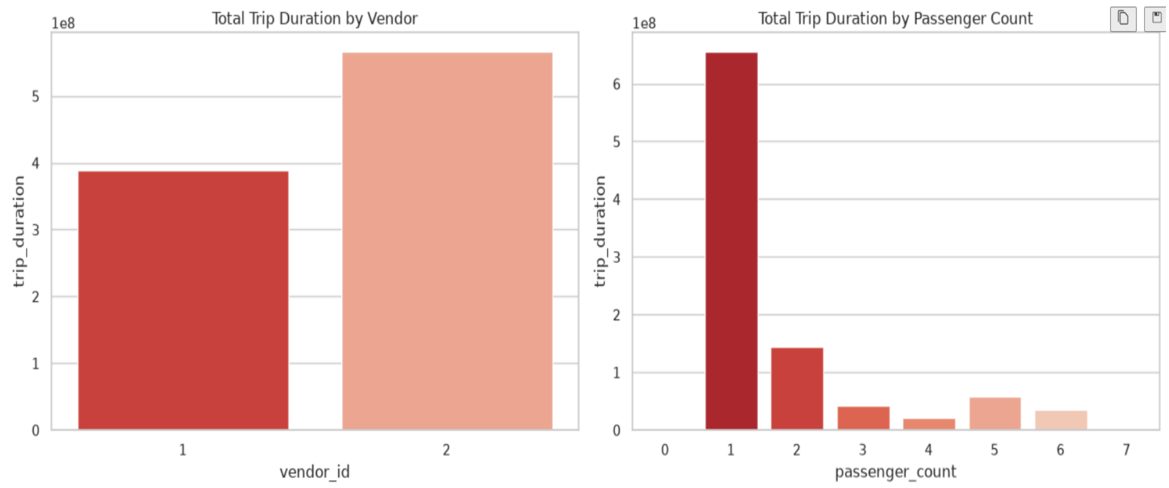


The target distribution look like gaussian distribution have right skew this aim to outliers in data.The shape distribution around the mean (6.5) = 665 sec



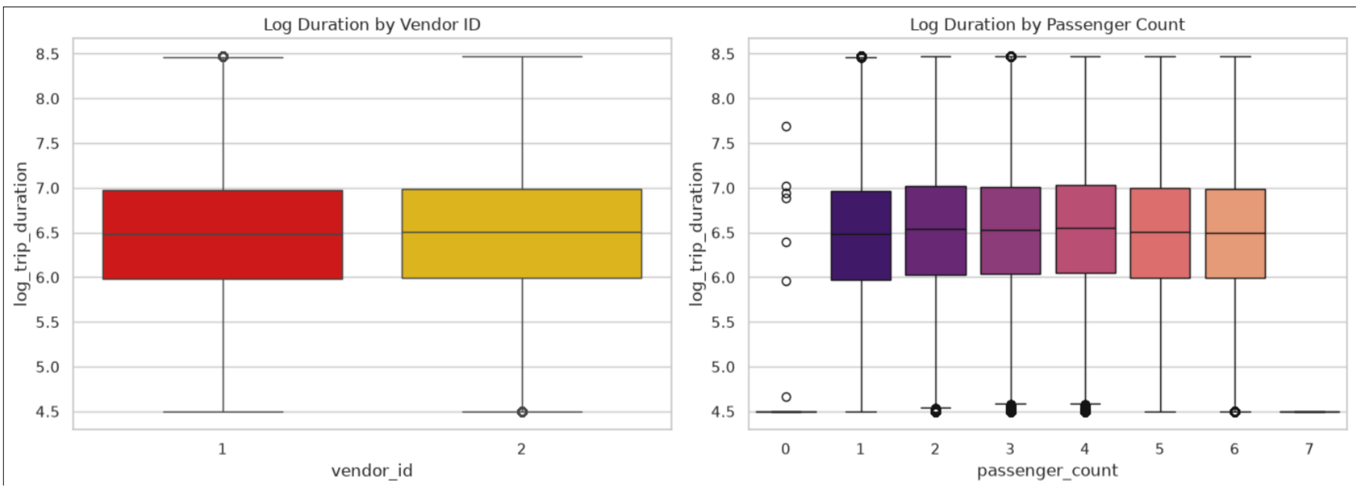
Applying clipping on data help me to reduce the effect to outliers this help model to good fit

2.2 Discreet Features



Vendor id 2 have more duration time than id 1 the can inside to some cases, vendor 2 have more trips then vendor 1

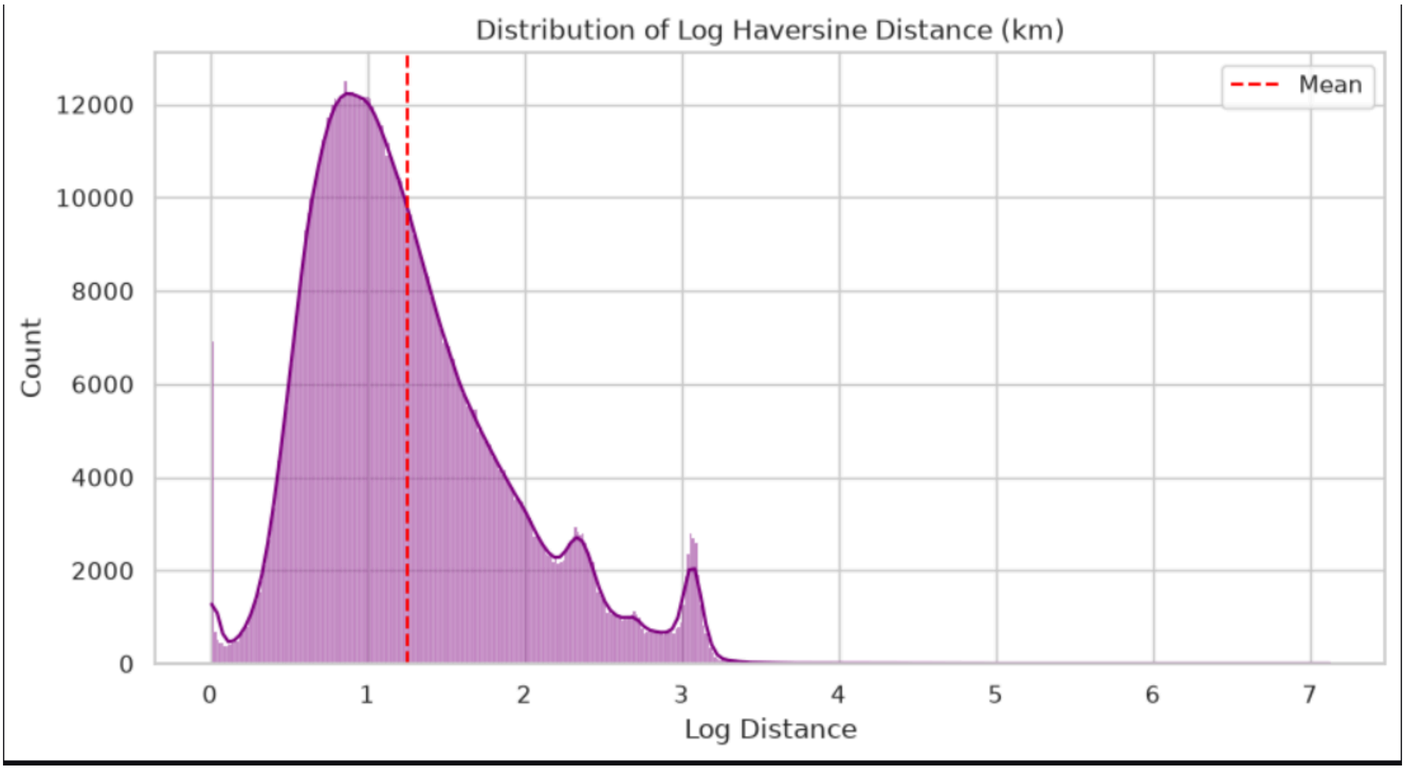
Passenger count 0,7 have low trip duration this may be outliers , passenger 1 have highest trip duration this insides to most trips to passenger 1



Vendor Distribution: Vendor 2 records slightly higher cumulative trip durations than vendor 1, but boxplots reveal that their median log trip durations are nearly identical.

Invalid Passenger Counts: Rides recorded with 0 passengers occur in the dataset. These are likely manual entry errors or freight deliveries, but they show similar duration distribution to 1-passenger trips.

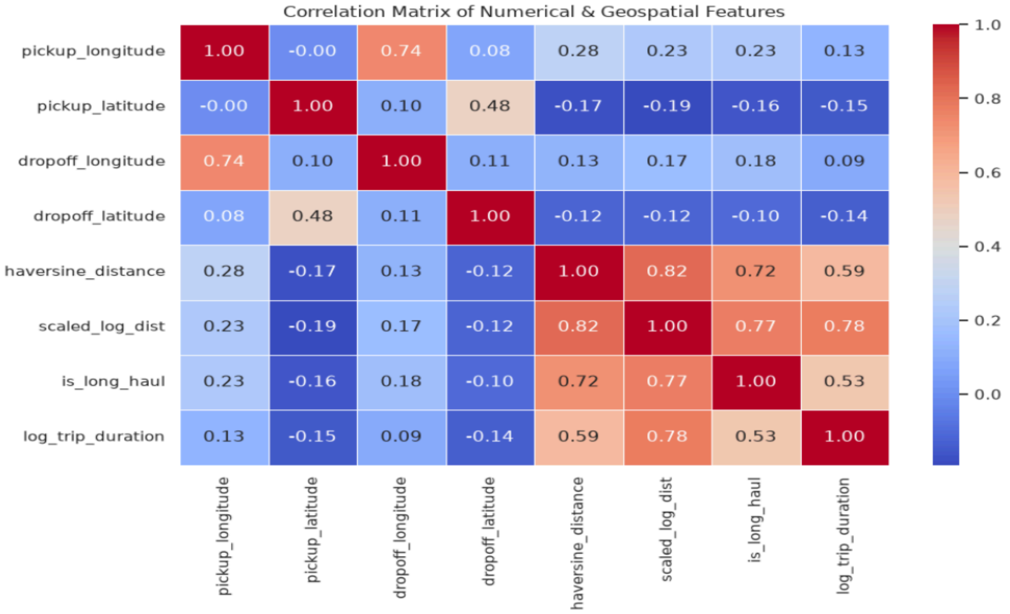
Passenger Distribution: Rides with 1 to 2 passengers dominate total volume. High passenger counts (7 passengers) are rare edge cases (likely large vans/SUVs) and contribute minimally to total dataset variance.



Distance Correlation: The great-circle (Haversine) distance derived from raw coordinates serves as a major physical predictor for trip_duration.

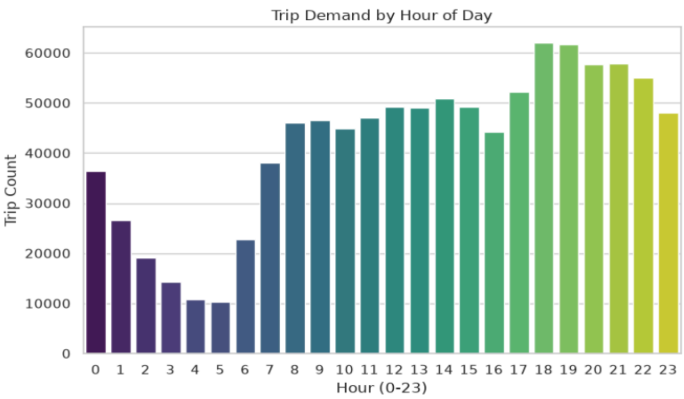
Zero-Distance Anomaly: Zero or near-zero distances represent trips where pickup and dropoff points are identical (e.g., canceled trips or round-trips).

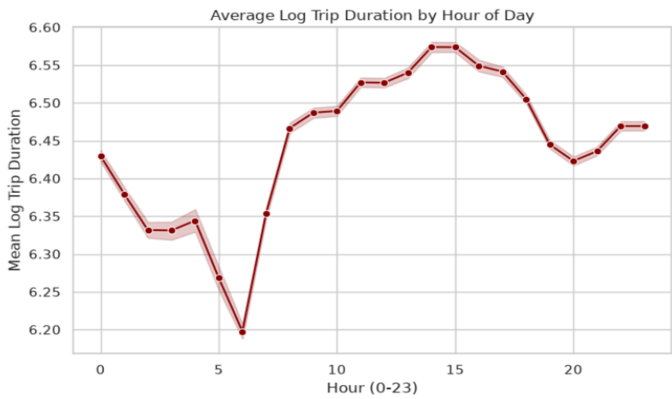
Clipping small values avoids mathematical errors during log transformation (log(0)).Scaling:
Applying log transformation and standard scaling to the distance metric addresses its strong right



Primary Drivers: scaled_log_dist, log_distance, and haversine_distance show the strongest positive correlation with log_trip_duration.

Geospatial Multicollinearity: Pickup and dropoff coordinates correlate strongly with each other, reinforcing why distance transformation and coordinate-relative features perform better than raw latitude/longitude inputs.

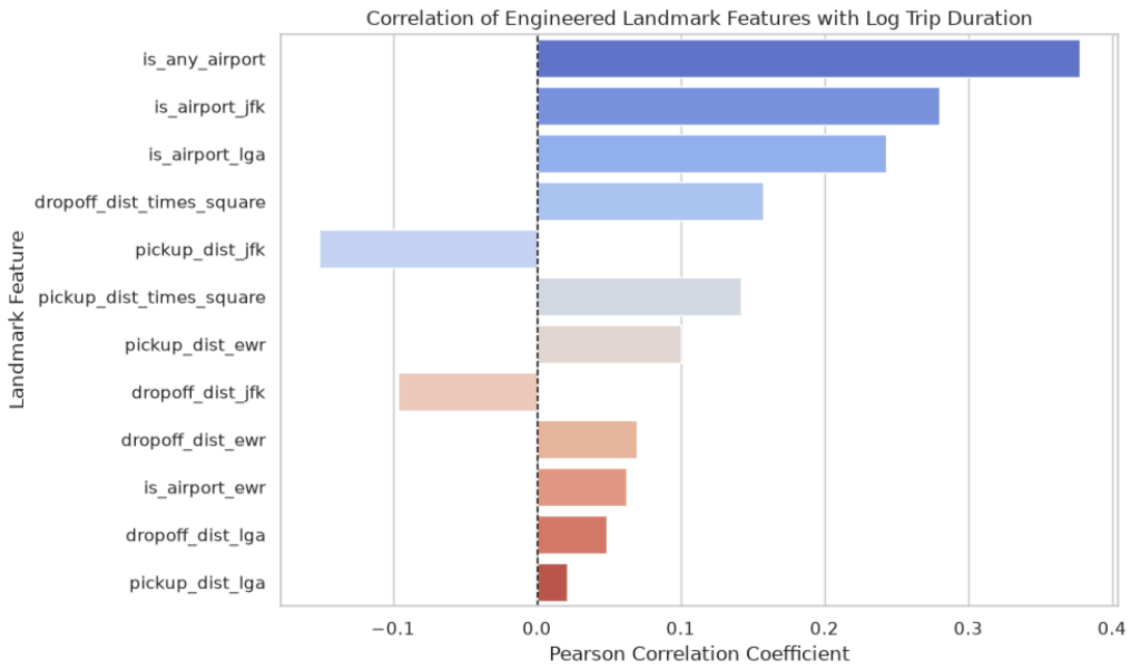




Hourly Demand Patterns: Trip volume peaks during evening rush hours (5 PM – 7 PM) and reaches its lowest point during early morning hours (3 AM – 5 AM).

Traffic Congestion Effect: Average log trip duration increases during daytime and rush hours despite potential speed drops, reflecting traffic delays.

Memory Flag: The store_and_fwd_flag indicates temporary connection drops before server transfer; mapping it to binary ($0/1$) makes it usable in modeling pipelines.



Strongest Predictive Drivers: pickup_dist_to_center and dropoff_dist_to_center typically exhibit strong correlations with log_trip_duration. Trips starting or ending farther from Midtown Manhattan often involve highway miles or long borough cross-town routes, directly extending total trip duration.

Airport Impact: Features like is_airport_jfk and is_any_airport show positive correlation spikes with trip duration due to long transit distances, heavy expressway congestion, and specific toll routes.

Symmetry: Pickup and dropoff distance metrics for the same landmark (e.g., pickup_dist_jfk vs. dropoff_dist_jfk) usually display mirrored correlation directions, giving non-linear models strong directional signals for traffic routing.

3. Modeling

We use the Ridge model to predict data pipeline we split data to numerical and categorical features and apply stander scaler on numerical and encoding on categorical and apply polynomial (degree=2)

3.1 Results:

```
r2score: 0.7160, mse: 0.1575
(nyc_env) marwan@DellLatitude30
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3.2 Lessons and future work:

Make the type of data version control is very beneficial for error analysis and verify assumption

Some insides :

1-the feathers selection improve performance

2-make deep analysis and connect some feats can help to make feat increase performance

3-some engineers add speed to data this make data leakage to model

4-using log transformation can increase the model performance make cost function more smooth to gradian decent

5-remove outliers can help model to make best fit

6-linear models often exhibit high bias. Given the characteristics of this datasets, it is important to explore more complex algorithms.For heterogeneous tabular data, techniques such as XGBoost and ensemble methods can be particularly effective.